

Creation Date 28-Apr-2009

Revision Date 15-May-2025

Revision Number 1

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                             |                    |
|-----------------------------|--------------------|
| <b>Product Description:</b> | <b>Bio-Acetone</b> |
| <b>Cat No. :</b>            | <b>S37187</b>      |
| <b>Synonyms</b>             | 2-Propanone        |
| <b>Index No</b>             | 606-001-00-8       |
| <b>CAS No</b>               | 67-64-1            |
| <b>EC No</b>                | 200-662-2          |
| <b>Molecular Formula</b>    | C3 H6 O            |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                             |                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>      | Laboratory chemicals.                                                                    |
| <b>Sector of use</b>        | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites |
| <b>Product category</b>     | PC21 - Laboratory chemicals                                                              |
| <b>Process categories</b>   | PROC15 - Use as a laboratory reagent                                                     |
| <b>Uses advised against</b> | No Information available                                                                 |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

ALFAAS37187

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Flammable liquids

Category 2 (H225)

### Health hazards

Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (single exposure)

Category 2 (H319)  
Category 3 (H336)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H225 - Highly flammable liquid and vapor  
H319 - Causes serious eye irritation  
H336 - May cause drowsiness or dizziness  
EUH066 - Repeated exposure may cause skin dryness or cracking

### **Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P280 - Wear eye protection/ face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P337 + P313 - If eye irritation persists: Get medical advice/attention

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)  
This product does not contain any known or suspected endocrine disruptors

## **Section 3: Composition/information on ingredients**

### 3.1. Substances

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| Component | CAS No  | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                        |
|-----------|---------|-----------|----------|--------------------------------------------------------------------------|
| Acetone   | 67-64-1 | 200-662-2 | >95      | Flam. Liq. 2 (H225)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H336)<br>EUH066 |

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Self-Protection of the First Aider</b> | Remove all sources of ignition. Use personal protective equipment as required.                                    |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: May cause pulmonary edema

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                                                 |
|---------------------------|-------------------------------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. Symptoms may be delayed. |
|---------------------------|-------------------------------------------------|

## Section 5: Firefighting measures

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

Do not use water jetstream.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Formaldehyde, Methanol.

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## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Flammables area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

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## 8.1. Control parameters

### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union                                        | The United Kingdom                                                                            | France                                                                                                                                                                                                                     | Belgium                                                                                                                         | Spain                                                                             |
|-----------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Acetone   | TWA: 500 ppm (8h)<br>TWA: 1210 mg/m <sup>3</sup> (8h) | TWA: 500 ppm<br>TWA: 1210 mg/m <sup>3</sup><br>STEL: 1500 ppm<br>STEL: 3620 mg/m <sup>3</sup> | TWA / VME: 500 ppm (8 heures). restrictive limit<br>TWA / VME: 1210 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 1000 ppm. restrictive limit<br>STEL / VLCT: 2420 mg/m <sup>3</sup> . restrictive limit | TWA: 246 ppm 8 uren<br>TWA: 594 mg/m <sup>3</sup> 8 uren<br>STEL: 492 ppm 15 minuten<br>STEL: 1187 mg/m <sup>3</sup> 15 minuten | TWA / VLA-ED: 500 ppm (8 horas)<br>TWA / VLA-ED: 1210 mg/m <sup>3</sup> (8 horas) |

| Component | Italy                                                                                                    | Germany                                     | Portugal                                                                                | The Netherlands                                                                                                                | Finland                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetone   | TWA: 500 ppm 8 ore.<br>Time Weighted Average<br>TWA: 1210 mg/m <sup>3</sup> 8 ore. Time Weighted Average | TWA: 500 ppm<br>TWA: 1200 mg/m <sup>3</sup> | STEL: 750 ppm 15 minutos<br>TWA: 500 ppm 8 horas<br>TWA: 1210 mg/m <sup>3</sup> 8 horas | STEL: 1 ppm 15 minuten<br>STEL: 2420 mg/m <sup>3</sup> 15 minuten<br>TWA: 500 ppm 8 uren<br>TWA: 1210 mg/m <sup>3</sup> 8 uren | TWA: 500 ppm 8 tunteina<br>TWA: 1200 mg/m <sup>3</sup> 8 tunteina<br>STEL: 630 ppm 15 minuutteina<br>STEL: 1500 mg/m <sup>3</sup> 15 minuutteina |

| Component | Austria                                                                                                                                                 | Denmark                                                                                                                             | Switzerland                                                                                                                             | Poland                                                                             | Norway                                                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetone   | MAK-KZGW: 2000 ppm 15 Minuten<br>MAK-KZGW: 4800 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 500 ppm 8 Stunden<br>MAK-TMW: 1200 mg/m <sup>3</sup> 8 Stunden | TWA: 250 ppm 8 timer<br>TWA: 600 mg/m <sup>3</sup> 8 timer<br>STEL: 500 ppm 15 minutter<br>STEL: 1200 mg/m <sup>3</sup> 15 minutter | STEL: 1000 ppm 15 Minuten<br>STEL: 2400 mg/m <sup>3</sup> 15 Minuten<br>TWA: 500 ppm 8 Stunden<br>TWA: 1200 mg/m <sup>3</sup> 8 Stunden | STEL: 1800 mg/m <sup>3</sup> 15 minutach<br>TWA: 600 mg/m <sup>3</sup> 8 godzinach | TWA: 125 ppm 8 timer<br>TWA: 295 mg/m <sup>3</sup> 8 timer<br>STEL: 156.25 ppm 15 minutter. value calculated<br>STEL: 368.75 mg/m <sup>3</sup> 15 minutter. value calculated |

| Component | Bulgaria                                                    | Croatia                                                                 | Ireland                                                                                                                 | Cyprus                                                                                 | Czech Republic                                                            |
|-----------|-------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Acetone   | TWA: 600 mg/m <sup>3</sup><br>STEL : 1400 mg/m <sup>3</sup> | TWA-GVI: 500 ppm 8 satima.<br>TWA-GVI: 1210 mg/m <sup>3</sup> 8 satima. | TWA: 500 ppm 8 hr.<br>TWA: 1210 mg/m <sup>3</sup> 8 hr.<br>STEL: 1500 ppm 15 min<br>STEL: 3630 mg/m <sup>3</sup> 15 min | Skin-potential for cutaneous absorption<br>TWA: 500 ppm<br>TWA: 1210 mg/m <sup>3</sup> | TWA: 800 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 1500 mg/m <sup>3</sup> |

| Component | Estonia                                                             | Gibraltar                                             | Greece                                                      | Hungary                                                               | Iceland                                                                                                                             |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Acetone   | TWA: 500 ppm 8 tundides.<br>TWA: 1210 mg/m <sup>3</sup> 8 tundides. | TWA: 500 ppm 8 hr<br>TWA: 1210 mg/m <sup>3</sup> 8 hr | STEL: 3560 mg/m <sup>3</sup><br>TWA: 1780 mg/m <sup>3</sup> | TWA: 500 ppm 8 órában. AK<br>TWA: 1210 mg/m <sup>3</sup> 8 órában. AK | TWA: 250 ppm 8 klukkustundum.<br>TWA: 600 mg/m <sup>3</sup> 8 klukkustundum.<br>Ceiling: 500 ppm<br>Ceiling: 1200 mg/m <sup>3</sup> |

| Component | Latvia                                      | Lithuania                                                                                               | Luxembourg                                                      | Malta                                       | Romania                                                 |
|-----------|---------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------|
| Acetone   | TWA: 500 ppm<br>TWA: 1210 mg/m <sup>3</sup> | TWA: 500 ppm IPRD<br>TWA: 1210 mg/m <sup>3</sup> IPRD<br>STEL: 1000 ppm<br>STEL: 2420 mg/m <sup>3</sup> | TWA: 500 ppm 8 Stunden<br>TWA: 1210 mg/m <sup>3</sup> 8 Stunden | TWA: 500 ppm<br>TWA: 1210 mg/m <sup>3</sup> | TWA: 500 ppm 8 ore<br>TWA: 1210 mg/m <sup>3</sup> 8 ore |

| Component | Russia                          | Slovak Republic | Slovenia            | Sweden               | Turkey              |
|-----------|---------------------------------|-----------------|---------------------|----------------------|---------------------|
| Acetone   | TWA: 200 mg/m <sup>3</sup> 1763 | TWA: 500 ppm    | TWA: 500 ppm 8 urah | Indicative STEL: 500 | TWA: 500 ppm 8 saat |

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|  |                            |                             |                                                                                                            |                                                                                                                                                |                                    |
|--|----------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
|  | MAC: 800 mg/m <sup>3</sup> | TWA: 1210 mg/m <sup>3</sup> | TWA: 1210 mg/m <sup>3</sup> 8 urah<br>STEL: 2420 mg/m <sup>3</sup> 15 minutah<br>STEL: 1000 ppm 15 minutah | ppm 15 minuter<br>Indicative STEL: 1200 mg/m <sup>3</sup> 15 minuter<br>TLV: 250 ppm 8 timmar. NGV<br>TLV: 600 mg/m <sup>3</sup> 8 timmar. NGV | TWA: 1210 mg/m <sup>3</sup> 8 saat |
|--|----------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                      | Spain                               | Germany                                |
|-----------|----------------|----------------|-----------------------------|-------------------------------------|----------------------------------------|
| Acetone   |                |                | Acetone: urine end of shift | Acetone: 50 mg/L urine end of shift | Acetone: 50 mg/L urine (end of shift ) |

| Component | Italy | Finland | Denmark | Bulgaria                                                           | Romania                             |
|-----------|-------|---------|---------|--------------------------------------------------------------------|-------------------------------------|
| Acetone   |       |         |         | Acetone: 80 mg/L urine at the end of exposure or end of work shift | Acetone: 50 mg/L urine end of shift |

| Component | Gibraltar | Latvia | Slovak Republic                                      | Luxembourg | Turkey |
|-----------|-----------|--------|------------------------------------------------------|------------|--------|
| Acetone   |           |        | Acetone: 80 mg/L urine end of exposure or work shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                  | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|----------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Acetone<br>67-64-1 ( >95 ) |                              |                                 |                                | DNEL = 186mg/kg bw/day            |

| Component                  | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|----------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Acetone<br>67-64-1 ( >95 ) | DNEL = 2420mg/m <sup>3</sup>     |                                     |                                    | DNEL = 1210mg/m <sup>3</sup>          |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                  | Fresh water     | Fresh water sediment         | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|----------------------------|-----------------|------------------------------|--------------------|------------------------------------|--------------------------|
| Acetone<br>67-64-1 ( >95 ) | PNEC = 10.6mg/L | PNEC = 30.4mg/kg sediment dw | PNEC = 21mg/L      | PNEC = 100mg/L                     | PNEC = 29.5mg/kg soil dw |

| Component | Marine water    | Marine water sediment | Marine water Intermittent | Food chain | Air |
|-----------|-----------------|-----------------------|---------------------------|------------|-----|
| Acetone   | PNEC = 1.06mg/L | PNEC = 3.04mg/kg      |                           |            |     |

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|               |  |             |  |  |  |
|---------------|--|-------------|--|--|--|
| 67-64-1 (>95) |  | sediment dw |  |  |  |
|---------------|--|-------------|--|--|--|

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material  | Breakthrough time | Glove thickness | EU standard    | Glove comments                                                                 |
|-----------------|-------------------|-----------------|----------------|--------------------------------------------------------------------------------|
| Butyl rubber    | > 480 minutes     | 0.5 mm          | EN 374 Level 6 | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene gloves | < 30 minutes      | 0.45 mm         |                |                                                                                |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Do not allow material to contaminate ground water system.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                                 |                       |                       |
|---------------------------------|-----------------------|-----------------------|
| <b>Physical State</b>           | Liquid                |                       |
| <b>Appearance</b>               | Colorless             |                       |
| <b>Odor</b>                     | sweet                 |                       |
| <b>Odor Threshold</b>           | 19.8 ppm              |                       |
| <b>Melting Point/Range</b>      | -95 °C / -139 °F      |                       |
| <b>Softening Point</b>          | No data available     |                       |
| <b>Boiling Point/Range</b>      | 56 °C / 132.8 °F      |                       |
| <b>Flammability (liquid)</b>    | Highly flammable      | On basis of test data |
| <b>Flammability (solid,gas)</b> | Not applicable        | Liquid                |
| <b>Explosion Limits</b>         | <b>Lower</b> 2.1 vol% |                       |

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|                                         |                                 |                          |
|-----------------------------------------|---------------------------------|--------------------------|
| Flash Point                             | Upper 13 vol%<br>-20 °C / -4 °F | Method - CC (closed cup) |
| Autoignition Temperature                | 465 °C / 869 °F                 |                          |
| Decomposition Temperature               | > 4°C                           |                          |
| pH                                      | 7                               |                          |
| Viscosity                               | 0.32 mPa.s @ 20 °C              |                          |
| Water Solubility                        | Soluble                         |                          |
| Solubility in other solvents            | No information available        |                          |
| Partition Coefficient (n-octanol/water) |                                 |                          |
| Component                               | log Pow                         |                          |
| Acetone                                 | -0.24                           |                          |
| Vapor Pressure                          | 247 mbar @ 20 °C                |                          |
| Density / Specific Gravity              | 0.790                           |                          |
| Bulk Density                            | Not applicable                  | Liquid                   |
| Vapor Density                           | 2.0                             | (Air = 1.0)              |
| Particle characteristics                | Not applicable (liquid)         |                          |

## 9.2. Other information

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| Molecular Formula    | C3 H6 O                                                   |
| Molecular Weight     | 58.08                                                     |
| Explosive Properties | Not explosive Vapors may form explosive mixtures with air |
| Oxidizing Properties | Not oxidising                                             |
| Evaporation Rate     | 5.6 (Butyl Acetate = 1.0)                                 |
| Refractive index     | 1.358 - 1.359                                             |

## Section 10: Stability and reactivity

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Heat, flames and sparks. Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong reducing agents. Strong bases. Peroxides. Halogenated compounds. Alkali metals. Amines.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Formaldehyde. Methanol.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information



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**(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

| Component | LD50 Oral          | LD50 Dermal                                  | LC50 Inhalation     |
|-----------|--------------------|----------------------------------------------|---------------------|
| Acetone   | 5800 mg/kg ( Rat ) | > 15800 mg/kg (rabbit)<br>> 7400 mg/kg (rat) | 76 mg/l, 4 h, (rat) |

**(b) skin corrosion/irritation;**

Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;**

**Test method**

Category 2

**Test species**

OECD 405

**Observation end point**

rabbit

Irritating to eyes

**(d) respiratory or skin sensitization;**

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

| Component                  | Test method                            | Test species | Study result    |
|----------------------------|----------------------------------------|--------------|-----------------|
| Acetone<br>67-64-1 ( >95 ) | Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | non-sensitising |

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

| Component                  | Test method                                                | Test species | Study result |
|----------------------------|------------------------------------------------------------|--------------|--------------|
| Acetone<br>67-64-1 ( >95 ) | OECD Test Guideline 471<br>AMES test                       | in vivo      | negative     |
|                            | OECD Test Guideline 476<br>Mammalian<br>Gene cell mutation | in vitro     | negative     |

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Based on available data, the classification criteria are not met

**(h) STOT-single exposure;**

Category 3

**Results / Target organs**

Central nervous system (CNS).

**(i) STOT-repeated exposure;**

Based on available data, the classification criteria are not met

**Test method**

OECD Test No. 408

**Test species / Duration**

Rat / 90 days

**Study result**

NOAEL = 900 mg/kg

**Route of exposure**

Oral

**Target Organs**

None known.

**(j) aspiration hazard;**

Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.  
May cause pulmonary edema.

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## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

#### Ecotoxicity effects

| Component | Freshwater Fish                                                                                                                                                         | Water Flea                                                             | Freshwater Algae              |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------|
| Acetone   | Oncorhynchus mykiss: LC50 = 5540 mg/l 96h<br>Alburnus alburnus: LC50 = 11000 mg/l 96h<br>Leuciscus idus: LC50 = 11300 mg/L/48h<br>Salmo gairdneri: LC50 = 6100 mg/L/24h | EC50 = 8800 mg/L/48h<br>EC50 = 12700 mg/L/48h<br>EC50 = 12600 mg/L/48h | NOEC = 430 mg/l (algae; 96 h) |

| Component | Microtox                 | M-Factor |
|-----------|--------------------------|----------|
| Acetone   | EC50 = 14500 mg/L/15 min |          |

### 12.2. Persistence and degradability

#### Persistence

Readily biodegradable  
Persistence is unlikely, based on information available.

| Component                | Degradability            |
|--------------------------|--------------------------|
| Acetone<br>67-64-1 (>95) | 91 % (28 d) (OECD 301 B) |

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Acetone   | -0.24   | 0.69 dimensionless            |

### 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

### 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant

This product does not contain any known or suspected substance

#### Ozone Depletion Potential

This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### 13.1. Waste treatment methods

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|                                            |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                   |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.                                                                                                    |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## Section 14: Transport information

### IMDG/IMO

|                                         |         |
|-----------------------------------------|---------|
| <b>14.1. UN number</b>                  | UN1090  |
| <b>14.2. UN proper shipping name</b>    | ACETONE |
| <b>14.3. Transport hazard class(es)</b> | 3       |
| <b>14.4. Packing group</b>              | II      |

### ADR

|                                         |         |
|-----------------------------------------|---------|
| <b>14.1. UN number</b>                  | UN1090  |
| <b>14.2. UN proper shipping name</b>    | ACETONE |
| <b>14.3. Transport hazard class(es)</b> | 3       |
| <b>14.4. Packing group</b>              | II      |

### IATA

|                                         |         |
|-----------------------------------------|---------|
| <b>14.1. UN number</b>                  | UN1090  |
| <b>14.2. UN proper shipping name</b>    | ACETONE |
| <b>14.3. Transport hazard class(es)</b> | 3       |
| <b>14.4. Packing group</b>              | II      |

|                                                                      |                                  |
|----------------------------------------------------------------------|----------------------------------|
| <b>14.5. Environmental hazards</b>                                   | No hazards identified            |
| <b>14.6. Special precautions for user</b>                            | No special precautions required. |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods   |

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

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| Component | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Acetone   | 67-64-1 | 200-662-2 | -      | -   | X     | X    | KE-29367 | X    | X    |

| Component | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Acetone   | 67-64-1 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Acetone   | 67-64-1 | -                                                                   | Use restricted. See entry 75. (see link for restriction details)              | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Acetone   | 67-64-1 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------|---------------------------------------|-------------------------|
| Acetone   | WGK1                                  |                         |

| Component | France - INRS (Tables of occupational diseases)      |
|-----------|------------------------------------------------------|
| Acetone   | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

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| Component                | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|--------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Acetone<br>67-64-1 (>95) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H225 - Highly flammable liquid and vapor  
H319 - Causes serious eye irritation  
H336 - May cause drowsiness or dizziness  
EUH066 - Repeated exposure may cause skin dryness or cracking

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

### Prepared By

Health, Safety and Environmental Department

### Creation Date

28-Apr-2009

### Revision Date

15-May-2025

### Revision Summary

Initial Release.

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**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**